

## Species Classification

- Carbohydrate Precursors (F07/F09)
- Amino Acid Precursors (F01/F06)
- Lipid Oxidation Precursors (F02)
- Reactive Intermediates / Amadori
- Desirable Volatiles (Aromas)
- Trapped Off-flavor Compounds
- Safety / Protein Damage Markers (F12)

## Reaction Families / Lanes & Intensities

- Schiff Base Formation
- Amadori Rearrangement
- Enolisation
- Sugar Dehydration
- Strecker Degradation
- Cysteine Degradation
- Thiol Addition
- Lipid Schiff Base
- Beta Elimination
- DHA Crosslinking
- High Intensity (Fast Michael addition)
- Medium-High (Fast Schiff/thiol add)
- Medium-Low (Moderate rearrangements)
- Low Intensity (Rate-limiting steps)
- Solid: Calibrated (Prior/Literature)
- Dashed: Heuristic (Uncalibrated)

